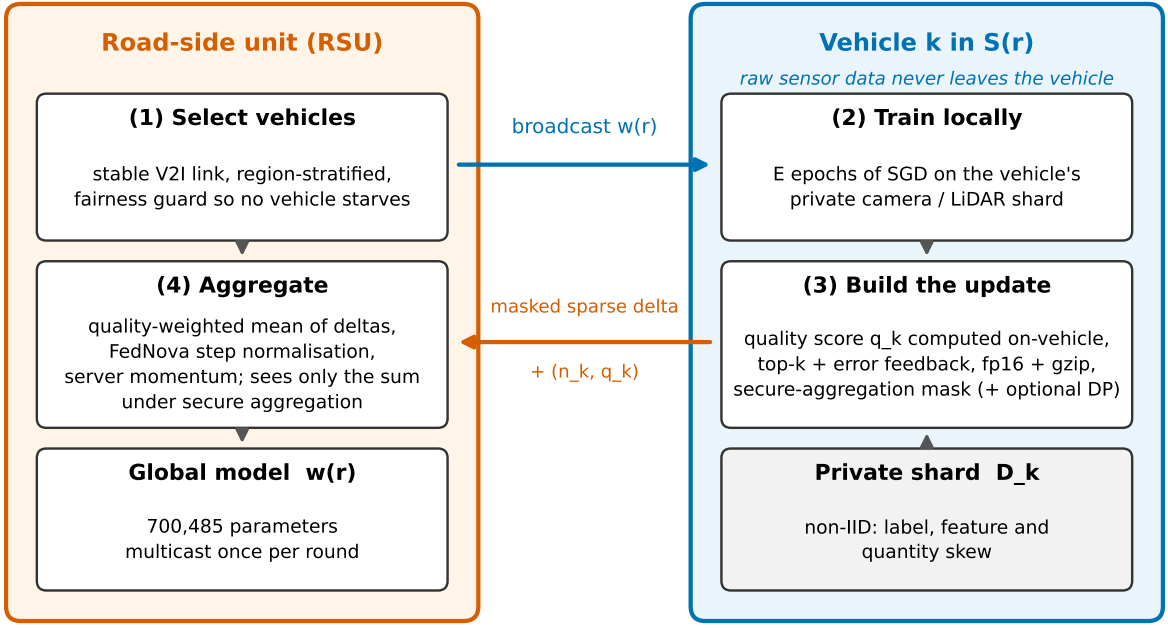
Road-side unit (RSU)
(1) Select vehicles
stable V2I link, region-stratified,
fairness guard so no vehicle starves
(4) Aggregate
quality-weighted mean of deltas,
FedNova step normalisation,
server momentum; sees only the sum
under secure aggregation
Global model w(r)
700,485 parameters
multicast once per round
broadcast w(r)
masked sparse delta
+ (n_k, q_k)
Vehicle k in S(r)
raw sensor data never leaves the vehicle
(2) Train locally
E epochs of SGD on the vehicle's
private camera / LiDAR shard
(3) Build the update
quality score q_k computed on-vehicle,
top-k + error feedback, fp16 + gzip,
secure-aggregation mask (+ optional DP)
Private shard D_k
non-IID: label, feature and
quantity skew